TATE MODULES OF ELLIPTIC CURVES

In this lecture, we study the ℓ-power torsion of an elliptic curve simultaneously, rather than one finite subgroup at a time. The resulting object is the *Tate module*. It turns isogenies into linear maps between free $\mathbb{Z}_\ell$-modules of rank two. Our main goal is to prove that this linear representation remains faithful even after allowing ℓ-adic coefficients in linear combinations of isogenies.

Throughout, K is a perfect field with fixed algebraic closure $\bar{K}$, and ℓ is a prime different from $\mathrm{char}(K)$. The elliptic curves are defined over K, but all torsion points are taken over $\bar{K}$.

1. The tower of torsion points

Recall from the previous lectures that the structure of $E[\ell^n]$ for an elliptic curve E is $\mathbb{Z}/\ell^n\mathbb{Z} \times \mathbb{Z}/\ell^n\mathbb{Z}$ as an abstract abelian group. A key number theoretic datum about the elliptic curve is the action of the Galois group G_K on this group. In particular, we can consider them all together via the following transition maps.

Proposition 1.1. *For every $n \geq 1$, the sequence*

$$0 \longrightarrow E[\ell] \longrightarrow E[\ell^{n+1}] \xrightarrow{[\ell]} E[\ell^n] \longrightarrow 0$$

is exact. Moreover, one can choose bases P_n, Q_n of $E[\ell^n]$ as a $\mathbb{Z}/\ell^n\mathbb{Z}$-module such that

$$[\ell]P_{n+1} = P_n, \qquad [\ell]Q_{n+1} = Q_n.$$

In particular, $E[\ell^n] \simeq (\mathbb{Z}/\ell^n\mathbb{Z})^2$.

Proof. This follos from standard facts and induction. □

2. The Tate module

Definition 2.1. The *ℓ-adic Tate module* of E is

$$T_\ell(E) = \varprojlim_n E[\ell^n],$$

where the transition maps are multiplication by ℓ. Explicitly,

$$T_\ell(E) = \{(R_n)_{n\geq 1} : R_n \in E[\ell^n],\ [\ell]R_{n+1} = R_n \text{ for all } n\}.$$

Addition is coordinatewise. For $a = (a_n) \in \mathbb{Z}_\ell$, define

$$a(R_n)_{n\geq 1} = ([a_n]R_n)_{n\geq 1}.$$

This is well-defined because R_n is killed by ℓ^n and the residues a_n are compatible. Thus $T_\ell(E)$ is a $\mathbb{Z}_\ell$-module.

Proposition 2.2. *The module $T_\ell(E)$ is free of rank two over $\mathbb{Z}_\ell$. The projection to the n-th coordinate induces a canonical isomorphism*

$$T_\ell(E)/\ell^n T_\ell(E) \xrightarrow{\sim} E[\ell^n]. \tag{2.1}$$

Proof. This is straightforward. □

Remark 2.3. The isomorphism $T_\ell(E) \simeq \mathbb{Z}_\ell^2$ is not canonical: it requires compatible bases. By contrast, (2.1) is canonical. The inverse-limit topology agrees with the ℓ-adic topology, whose neighbourhoods of zero are $\ell^n T_\ell(E)$. Every $\mathbb{Z}_\ell$-linear map between Tate modules is continuous, since it sends $\ell^n T_\ell(E_1)$ into $\ell^n T_\ell(E_2)$.

Definition 2.4. The *rational Tate module* is

$$V_\ell(E) = T_\ell(E) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

It is a two-dimensional vector space over $\mathbb{Q}_\ell$.

Example 2.5 (A rank-one analogue)**.** Let $\mu_{\ell^n} \subset \bar{K}^\times$ be the group of ℓ^n-th roots of unity. Then

$$\mathbb{Z}_\ell(1) := \varprojlim_n \mu_{\ell^n}, \qquad \zeta \longmapsto \zeta^\ell,$$

is a free $\mathbb{Z}_\ell$-module of rank one, written multiplicatively. A compatible sequence of primitive roots ζ_n gives the isomorphism

$$\mathbb{Z}_\ell \longrightarrow \mathbb{Z}_\ell(1), \qquad (a_n) \longmapsto (\zeta_n^{a_n}).$$

The choice of roots is analogous to the choice of a compatible basis for $T_\ell(E)$. We will encounter this rank-one module again with the Weil pairing.

3. Isogenies acting on Tate modules

Let $\phi \in \mathrm{Hom}(E_1, E_2)$. Since ϕ commutes with multiplication by integers, it maps $E_1[\ell^n]$ into $E_2[\ell^n]$ and induces

$$T_\ell(\phi) : T_\ell(E_1) \longrightarrow T_\ell(E_2), \qquad (R_n) \longmapsto (\phi(R_n)).$$

This map is $\mathbb{Z}_\ell$-linear. Directly from the definition,

$$\begin{aligned} T_\ell(\phi + \psi) &= T_\ell(\phi) + T_\ell(\psi), \\ T_\ell(\lambda \circ \phi) &= T_\ell(\lambda) \circ T_\ell(\phi), \\ T_\ell([m]) &= m \ \mathrm{id}\,. \end{aligned}$$

Thus the Tate module is *covariant*: it preserves the direction of a morphism, unlike pullback on functions or divisor classes.

Proposition 3.1. *If $\phi : E_1 \to E_2$ is a nonzero isogeny of degree d, then $T_\ell(\phi)$ is injective with finite cokernel. It induces an isomorphism*

$$V_\ell(\phi) : V_\ell(E_1) \xrightarrow{\sim} V_\ell(E_2).$$

If $\ell \nmid d$, then $T_\ell(\phi)$ itself is an isomorphism.

Proof. The dual-isogeny identities give

$$T_\ell(\widehat{\phi})T_\ell(\phi) = d \ \mathrm{id}, \qquad T_\ell(\phi)T_\ell(\widehat{\phi}) = d \ \mathrm{id}\,.$$

Since $d \neq 0$ in $\mathbb{Z}_\ell$ and $T_\ell(E_1)$ is torsion-free, the first identity proves injectivity. The second says that the cokernel is killed by d. It is therefore a quotient of $T_\ell(E_2)/dT_\ell(E_2)$, which is finite. After tensoring with $\mathbb{Q}_\ell$, an inverse is $d^{-1}T_\ell(\widehat{\phi})$. The same inverse is integral when d is a unit in $\mathbb{Z}_\ell$. □

Example 3.2. Multiplication by ℓ acts on $T_\ell(E)$ as ℓI_2. It is injective but not surjective, with cokernel $(\mathbb{Z}/\ell\mathbb{Z})^2$. This is quite different from $[\ell] : E(\bar{K}) \to E(\bar{K})$, which is surjective.

In characteristic $p > 0$, relative Frobenius $F : E \to E^{(p)}$ has degree p, and therefore induces an isomorphism on T_ℓ for $\ell \neq p$. Thus a purely inseparable isogeny is still detected by the prime-to-p Tate module, even though its pullback on regular differentials is zero.

Proposition 3.3. *The homomorphism*

$$\mathrm{Hom}(E_1, E_2) \longrightarrow \mathrm{Hom}_{\mathbb{Z}_\ell}\big(T_\ell(E_1), T_\ell(E_2)\big), \qquad \phi \longmapsto T_\ell(\phi),$$

is injective.

Proof. This is an exercise. □

Our main theorem will assert more: the corresponding map remains injective after replacing integer coefficients in $\mathrm{Hom}(E_1, E_2)$ by ℓ-adic coefficients. This requires an additional argument.

4. Vanishing on torsion and divisibility of homomorphisms

Put $H = \mathrm{Hom}(E_1, E_2)$. Multiplication by an integer m on this abelian group means

$$m\psi = [m]_{E_2} \circ \psi = \psi \circ [m]_{E_1}.$$

The following lemma is the main geometric input to the proof.

Lemma 4.1 (Divisibility detected by torsion)**.** *Let $m \geq 1$ be an integer with $m \neq 0$ in K, and let $\phi \in H$. Then*

$$\phi|_{E_1[m]} = 0 \quad \Longleftrightarrow \quad \phi \in mH.$$

More precisely, in this case there is a unique $\psi \in H$ such that

$$\phi = \psi \circ [m]_{E_1} = m\psi.$$

If ϕ is defined over K, then so is ψ.

Proof. If $\phi = m\psi$, then for $T \in E_1[m]$ we have $\phi(T) = \psi([m]T) = O_2$.

Conversely, suppose that ϕ kills $E_1[m]$. For every $T \in E_1[m]$,

$$\phi(P + T) = \phi(P) + \phi(T) = \phi(P).$$

Thus ϕ is invariant under translation by $E_1[m]$. We spell out why this invariance gives a morphism on the quotient.

Let $L = \bar{K}(E_1)$, let $\pi = [m]_{E_1}$, and let $G = E_1[m]$ act on L by translation. Since $\pi \circ \tau_T = \pi$, we have

$$\pi^* L \subseteq L^G.$$

The map π is separable of degree m^2, while G is a faithful group of m^2 automorphisms. The fixed-field theorem and the degree formula give

$$[L : \pi^* L] = m^2 = [L : L^G], \qquad \text{hence} \qquad \pi^* L = L^G.$$

If $\phi \neq 0$, its invariance gives

$$\phi^* \bar{K}(E_2) \subseteq L^G = \pi^* L.$$

Therefore $(\pi^*)^{-1} \circ \phi^*$ is an embedding $\bar{K}(E_2) \hookrightarrow L$. The correspondence between function fields and smooth projective curves gives a morphism $\psi : E_1 \to E_2$ satisfying $\phi = \psi \circ \pi$. Evaluating at O_1 gives $\psi(O_1) = O_2$, so $\psi \in H$. If $\phi = 0$, simply take $\psi = 0$.

Uniqueness follows from the surjectivity of π. Finally, if ϕ is defined over K, then $\psi^\sigma \circ \pi = \phi$ for every $\sigma \in G_K$. Uniqueness gives $\psi^\sigma = \psi$, so ψ is defined over K. □

Remark 4.2. Only $[m]$ needs to be separable here; ϕ need not be separable. This is exactly where the assumption $\ell \neq \mathrm{char}(K)$ will enter when $m = \ell^n$.

Corollary 4.3. *For every $n \geq 1$, restriction to torsion induces an injection*

$$H/\ell^n H \hookrightarrow \mathrm{Hom}_{\mathbb{Z}/\ell^n\mathbb{Z}}\big(E_1[\ell^n], E_2[\ell^n]\big). \tag{4.1}$$

In particular, $H/\ell^n H$ is finite, with at most ℓ^{4n} elements.

Proof. Restriction is an additive map on H, and Lemma 4.1 says that its kernel is exactly $\ell^n H$. Choosing bases of the two torsion groups identifies the target with $M_2(\mathbb{Z}/\ell^n\mathbb{Z})$, a set of ℓ^{4n} elements. □

5. Finite generation of H

We shall need to know that H is finitely generated. It would be circular to deduce this from the main Tate-module theorem and then use it in that theorem's proof. Instead, the degree function supplies a direct argument.

Write $q(\phi) = \deg \phi$, with $q(0) = 0$. From the preceding lecture,

$$q(\phi) \in \mathbb{Z}_{\geq 0}, \qquad q(\phi) = 0 \iff \phi = 0,$$
$$q(m\phi) = m^2 q(\phi),$$
$$q(\phi + \psi) + q(\phi - \psi) = 2q(\phi) + 2q(\psi).$$

Since degrees are nonnegative, the last equality implies the useful bound

$$q(\phi - \psi) \le 2q(\phi) + 2q(\psi). \tag{5.1}$$

Lemma 5.1 (There are finitely many homomorphisms of bounded degree)**.** *For every real number $B \ge 0$, the set*

$$H_{\le B} = \{\phi \in H : q(\phi) \le B\}$$

is finite.

Proof. Choose n such that $\ell^{2n} > 4B$. Suppose that $\phi, \psi \in H_{\le B}$ have the same restriction to $E_1[\ell^n]$. By Lemma 4.1,

$$\phi - \psi = \ell^n \theta$$

for some $\theta \in H$. If $\theta \neq 0$, then

$$q(\phi - \psi) = \ell^{2n} q(\theta) \ge \ell^{2n} > 4B.$$

But (5.1) gives $q(\phi - \psi) \le 4B$, a contradiction. Thus $\phi = \psi$. Restriction therefore injects $H_{\le B}$ into the finite set $\mathrm{Hom}(E_1[\ell^n], E_2[\ell^n])$. □

Proposition 5.2. *The abelian group $H = \mathrm{Hom}(E_1, E_2)$ is finitely generated and torsion-free. Consequently, it is a free abelian group of finite rank.*

Proof. By Corollary 4.3, choose a finite set $R \subset H$ of representatives for $H/\ell H$, including 0, and put

$$C = \max_{r \in R} q(r).$$

The set $S = H_{\le C}$ is finite by Lemma 5.1, and it contains R. We show that S generates H by induction on the nonnegative integer $q(\phi)$.

If $q(\phi) \le C$, this is immediate. Otherwise, write

$$\phi = r + \ell\psi, \qquad r \in R,\ \psi \in H.$$

Using (5.1) and $\ell \ge 2$, we obtain

$$q(\psi) = \frac{q(\phi - r)}{\ell^2} \le \frac{2q(\phi) + 2C}{\ell^2} \le \frac{q(\phi) + C}{2} < q(\phi).$$

The induction hypothesis expresses ψ as an integer combination of elements of S, and hence does the same for ϕ. This proves finite generation.

If $m\phi = 0$ for a nonzero integer m, then $0 = q(m\phi) = m^2 q(\phi)$, so $\phi = 0$. Thus H is torsion-free. The structure theorem for finitely generated abelian groups finishes the proof. □

Notice what made the argument work: finite torsion detects a bounded-degree homomorphism, and division by ℓ after subtracting one of finitely many representatives lowers the degree. No assertion about the rank of H was used.

6. The integral injectivity theorem

The additive action of H on Tate modules extends to a $\mathbb{Z}_\ell$-linear map

$$\begin{aligned} \Theta_\ell : H \otimes_{\mathbb{Z}} \mathbb{Z}_\ell &\longrightarrow \mathrm{Hom}_{\mathbb{Z}_\ell}\big(T_\ell(E_1), T_\ell(E_2)\big), \\ \phi \otimes a &\longmapsto a\, T_\ell(\phi). \end{aligned}$$

The tensor product is over $\mathbb{Z}$. Its elements are finite sums of tensors; after choosing a $\mathbb{Z}$-basis of H, they are linear combinations of that basis with coefficients in $\mathbb{Z}_\ell$.

Theorem 6.1 (Faithfulness with ℓ-adic coefficients)**.** *Let E_1, E_2 be elliptic curves, and let $\ell \neq \mathrm{char}(K)$ be a prime. Then the natural map*

$$\boxed{\mathrm{Hom}(E_1, E_2) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \hookrightarrow \mathrm{Hom}_{\mathbb{Z}_\ell}\big(T_\ell(E_1), T_\ell(E_2)\big)}$$

is injective.

Proof. By Proposition 5.2, choose a $\mathbb{Z}$-basis $\phi_1, \dots, \phi_r$ of H. The case $r = 0$ is immediate. Suppose that

$$\alpha = \sum_{i=1}^{r} \phi_i \otimes a_i, \qquad a_i \in \mathbb{Z}_\ell,$$

has zero image under Θ_ℓ. Thus

$$\sum_{i=1}^{r} a_i T_\ell(\phi_i) = 0 \quad \text{as a map } T_\ell(E_1) \to T_\ell(E_2). \tag{6.1}$$

We must prove that every a_i is zero.

Fix $n \geq 1$. Choose integers b_i such that

$$b_i \equiv a_i \pmod{\ell^n \mathbb{Z}_\ell},$$

and form the actual homomorphism of elliptic curves

$$f_n = \sum_{i=1}^{r} b_i \phi_i \in H.$$

Reducing (6.1) modulo ℓ^n gives

$$\sum_{i=1}^{r} (b_i \bmod \ell^n)\, \phi_i|_{E_1[\ell^n]} = 0,$$

where we used the canonical identifications

$$T_\ell(E_j)/\ell^n T_\ell(E_j) \simeq E_j[\ell^n].$$

In other words, f_n kills $E_1[\ell^n]$. Lemma 4.1 therefore gives

$$f_n = \ell^n \psi_n \qquad \text{for some } \psi_n \in H.$$

Write $\psi_n = \sum_i c_i \phi_i$ with $c_i \in \mathbb{Z}$. Comparing coefficients in the $\mathbb{Z}$-basis $\phi_1, \dots, \phi_r$ gives

$$b_i = \ell^n c_i \qquad (1 \leq i \leq r).$$

Thus $b_i \in \ell^n \mathbb{Z}$, and the congruence defining b_i implies

$$a_i \in \ell^n \mathbb{Z}_\ell \qquad (1 \leq i \leq r).$$

Since n was arbitrary and $\bigcap_{n \geq 1} \ell^n \mathbb{Z}_\ell = \{0\}$, every $a_i = 0$. Hence $\alpha = 0$, proving injectivity. □

Remark 6.2 (The same proof in inverse-limit language). Because H is finitely generated and free, there is a canonical isomorphism

$$H \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \simeq \varprojlim_n H/\ell^n H.$$

Indeed, after choosing a basis, both sides are $\mathbb{Z}_\ell^r$, and the canonical map is coefficientwise reduction. Similarly, if

$$A = \mathrm{Hom}_{\mathbb{Z}_\ell}\big(T_\ell(E_1), T_\ell(E_2)\big),$$

then choosing compatible bases gives

$$A/\ell^n A \simeq \mathrm{Hom}_{\mathbb{Z}/\ell^n\mathbb{Z}}\big(E_1[\ell^n], E_2[\ell^n]\big), \qquad A \simeq \varprojlim_n A/\ell^n A.$$

The finite-level injections (4.1) are compatible with reduction. Taking their inverse limit therefore gives the theorem: an element mapping to zero has every finite-level component zero. The coordinate proof above spells out this argument without assuming any exactness theorem for inverse limits.

Remark 6.3 (Why the easier injectivity statement is not enough). One cannot obtain Theorem 6.1 merely by saying that Proposition 3.3 stays injective after allowing ℓ-adic coefficients. For example, choose $c \in \mathbb{Z}_\ell \setminus \mathbb{Q}$. The map

$$\mathbb{Z}^2 \longrightarrow \mathbb{Z}_\ell, \qquad (a, b) \longmapsto a + cb,$$

is injective, but its $\mathbb{Z}_\ell$-linear extension

$$\mathbb{Z}_\ell^2 \longrightarrow \mathbb{Z}_\ell$$

kills the nonzero vector $(-c, 1)$. In our situation, the divisibility lemma rules out precisely this possibility. Finite generation also justifies identifying the tensor product with the inverse limit; that identification should not be assumed for an arbitrary abelian group.

Corollary 6.4. *The group* $\mathrm{Hom}(E_1, E_2)$ *is free of rank at most four. In particular,* $\mathrm{End}(E)$ *is a free abelian group of rank at most four, and, after choosing a basis of* $T_\ell(E)$*, there is an injective homomorphism of* $\mathbb{Z}_\ell$*-algebras*

$$\mathrm{End}(E) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longhookrightarrow M_2(\mathbb{Z}_\ell).$$

Proof. The target in Theorem 6.1 is free of rank four. If H has $\mathbb{Z}$-rank r, its tensor product with $\mathbb{Z}_\ell$ is free of rank r. Tensoring the injection with $\mathbb{Q}_\ell$ gives $r \le 4$. Compatibility with composition proves the algebra assertion for endomorphisms. □

7. The Galois action

We now take account of the field of definition. The group $G_K = \mathrm{Gal}(\bar{K}/K)$ acts on every $E[\ell^n]$, and this action commutes with multiplication by ℓ. Consequently,

$$\sigma(R_n)_{n\ge 1} = (\sigma R_n)_{n\ge 1}$$

defines an action on $T_\ell(E)$ by $\mathbb{Z}_\ell$-linear automorphisms. Choosing a basis gives the *ℓ-adic Galois representation*

$$\rho_{E,\ell} : G_K \longrightarrow \mathrm{Aut}_{\mathbb{Z}_\ell}(T_\ell(E)) \simeq \mathrm{GL}_2(\mathbb{Z}_\ell).$$

A different basis conjugates the matrices, so the representation is intrinsically an action on $T_\ell(E)$, not a preferred list of matrices.

Proposition 7.1. *The representation* $\rho_{E,\ell}$ *is continuous. If* $\phi : E_1 \to E_2$ *is defined over* K*, then* $T_\ell(\phi)$ *commutes with the* G_K*-actions.*

Proof. For each n, the finitely many coordinates of the points of $E[\ell^n]$ lie in a finite extension of K. Thus the action on $E[\ell^n]$ has an open kernel in G_K. These are the reductions of $\rho_{E,\ell}$ modulo ℓ^n, which proves continuity. The second assertion follows coordinatewise from $\phi(\sigma P) = \sigma\phi(P)$. □

Corollary 7.2. *There is an injection*

$$\mathrm{Hom}_K(E_1, E_2) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \longhookrightarrow \mathrm{Hom}_{\mathbb{Z}_\ell[G_K]}\big(T_\ell(E_1), T_\ell(E_2)\big).$$

Here the notation on the right means $\mathbb{Z}_\ell$*-linear maps commuting with every* $\sigma \in G_K$*.*

Proof. The group $H_K = \mathrm{Hom}_K(E_1, E_2)$ is a subgroup of the finite free group H, and hence is finite free. The last assertion of Lemma 4.1 shows that a K-defined homomorphism killing $E_1[\ell^n]$ belongs to $\ell^n H_K$, not merely to $\ell^n H$. The proof of Theorem 6.1 therefore applies to H_K as well. Galois equivariance puts its image in the indicated target. □

Remark 7.3. The geometric group $\mathrm{Hom}(E_1, E_2)$ and the group $\mathrm{Hom}_K(E_1, E_2)$ must be distinguished. A geometric homomorphism need not commute with the full G_K-action. Surjectivity onto Galois-equivariant homomorphisms is a different and much deeper question.